Forecast Accuracy, Decision Leverage, and Objective Mismatch in World-Model Planning

Research manuscript

16 October 2026

Abstract

Evaluating a world model by forecast error alone leaves two decision-level questions unanswered: how much can the evaluated selection interface change real outcomes, and would its scorer choose well even with actual future observations? We operationalize these questions with a finite-support utility audit and an oracle-feature scoring diagnostic. On 30 recorded Cosmos Policy development roots with two first-action continuations each, any selector over that fixed support attains between 28 and 30 successes. Relative to the recorded 29-success controller, first-choice-only damage is at most one additional failure, although completion cost retains more variation. A prospective DINO-WM reproduction experiment evaluates an added ranker over five frozen development action sequences on 30 independently reset MetaWorld reach roots. Replacing predicted features with actual-future encodings lowers mean 30-control terminal reward from 6.300613 to 4.758220: paired mean change -1.542393, with a 95% bootstrap interval of $[-2.438788, -0.715561]$. Reward decreases at 11 roots, ties at 18, and improves at one. Oracle forecast error is zero by construction, a manipulation check rather than a deployable forecasting improvement. Fixed choices of candidate0 and candidate2 achieve higher mean terminal rewards, 7.306371 and 7.262187, than the original ranker. A separate development example shows an oracle-feature ordering that no nonnegative, nonzero reweighting of the scorer's two components repairs. Exploratory reuse of the 30 roots with a frozen, development-fitted additive bias also lowers forecast MSE and selected utility. In a separate single-root native CEM query, the same correction lowers 15-control reward and return, but improves both over the returned mean's 30-control open-loop continuation. A complete native replanning episode at that same root raises 99-control return from 747.432862 to 760.171563, failing the preset persistent-loss gate, although first success is delayed from control 72 to 97. The evidence supports interface-specific diagnoses of a known prediction–control mismatch, with explicit baseline-competence and scope limitations; it does not establish an attack, a validated defense, or cross-task generality.

1 Introduction

World-model planning inserts a prediction between an action proposal and an action choice. An evaluation can consequently fail in several distinct ways: the available actions have nearly identical task outcomes; predicted futures are inaccurate; the scorer rewards the wrong aspects of even an accurate future; or a change to the prediction also changes which actions are proposed. Forecast divergence and action-rank changes do not identify which of these explanations applies. Actual environment outcomes and an explicit decision interface are necessary to separate them.

The distinction between prediction accuracy and decision quality is established in decision-focused learning and model-based reinforcement learning, including Lambert et al.'s objective-mismatch study [10]. We do not propose "better prediction, worse control" as a new discovery.

Our question is operational: what can be established from the interfaces and real continuation caches of existing pretrained world-model systems, without retraining a large model or labeling every changed forecast as a successful attack?

We study two complementary diagnostics. First, a complete finite candidate continuation table bounds the outcomes reachable by a selector restricted to that table. The useful quantity for harm is not just the range of outcomes, but the downside relative to the particular baseline being evaluated. Second, feeding the measured future representation to the same scorer tests whether prediction perfection on that bank would remove its regret. This substitution does not make the representation, objective, or candidate set perfect.

The results are intentionally tied to measured interfaces. The Cosmos audit contains 60 real continuations at 30 development roots. The prospective DINO diagnostic evaluates the same five frozen development action sequences at each of 30 fresh, independently seeded roots, using an added fixed-bank ranker. Earlier two-mean and five-action development diagnostics retain their exploratory status and are not pooled with the confirmation panel. A subsequently designed fixed-bias correction reuses the 30 roots and is therefore exploratory; a separate single-root native CEM comparison tests its execution-prefix effect and exposes a reversal under longer open-loop execution. A complete native episode then reuses that inspected root and finds higher cumulative return with the correction; it is a further exploratory check, not another independent root. WorldAgen's corrected short replays are an interface control because its frozen native policy does not rank world-head forecasts. Resource availability in additional systems is not empirical coverage of these tests.

Our contributions are a reproducible finite-cache evaluation procedure, an explicit account of the assumptions needed to connect it to a planner, and two measured diagnoses that alter the interpretation of earlier experiments, including an independent-root confirmation of an offline oracle-substitution effect within a prespecified fixed bank. The bounds and regret identity are elementary tools, not claimed new theorems. The substantive empirical question—how commonly prediction improvements decrease task utility in independently sampled native planning queries—remains open: the oracle confirmation does not regenerate native CEM proposals, the deployable-form correction reuses examined roots, and the native follow-ups cover only one root, with opposite short-prefix and complete-episode utility directions and no full-episode forecast-fidelity audit.

2 Related work and contribution boundary

Prediction quality and decision quality. Lambert et al. [10] identify objective mismatch in model-based reinforcement learning and empirically show that one-step prediction likelihood need not track downstream control performance. Consequently, better prediction together with worse control is not a new principle or discovery of this study. Smart Predict-then-Optimize measures errors through the decisions they induce rather than through a prediction loss alone [4]. Value-aware model learning and value equivalence adapt environment models to their downstream value calculations [5, 6]. TEMPO incorporates task-aware sample weighting into model training [14]. These precedents rule out treating the general distinction between predictive accuracy and control utility as our novelty. We instead operationalize a finite-support audit for frozen interfaces and existing branch-replay artifacts, then examine a fixed additive forecast correction and its horizon-dependent effect in a native CEM query.

Objective proxies. Pan et al. [12] study cases in which stronger optimization of a proxy accompanies lower true reward. Our true-future comparison concerns a frozen planning proxy and

a specified candidate bank, and does not establish deliberate reward hacking. The residual regret in Eq. (3) also includes representation and selection limitations. Adding and subtracting an oracle-interface utility is an accounting identity, rather than a new causal identification theorem.

Model exploitation and security claims. Bhamidipaty et al. [2] define model exploitation through opposite policy orderings under model and environment dynamics (Section 2.2), and explicitly discuss an optimizer producing such behavior without an external adversary (Section 1). This is close prior work for interpreting ordering disagreements. Our goal-feature scorer is not itself an estimate of task return, so proxy disagreement should not be conflated with their transition-model definition. Neither an ordering reversal nor the generic prediction–control mismatch establishes a new security mechanism. Such a claim would additionally need a specified feasible external intervention and measured real-utility consequences under that intervention.

Visual world-model planning. DINO-WM plans toward goal features using an action-conditioned latent predictor [15]. Cosmos Policy jointly represents actions, future observations, and values, and uses their predictions for planning [9]. Their different consumers motivate checking the actual action-selection interface before interpreting forecast changes. CF-JEPA separates controllable and uncontrollable latent components, whereas Dueling World Models centers predictions across actions to expose an action-dependent channel [3, 11].

The publicly available LENS submission abstract describes a frozen-model planning metric derived from real/predicted action-response agreement and prediction accuracy, including a fixed-candidate, fixed-prediction ranking comparison [1]. This directly overlaps the idea of reweighting latent directions to improve action rankings. We verified its official abstract and submission metadata, but could not access its full text; its reported performance is not evidence from our experiments. Consequently, we make no priority claim for latent reweighting, forecast versus decision evaluation, or oracle subtraction. The contribution sought here is a reproducible interface audit with explicit support, replay provenance, and claim-specific evidence. Its empirical scope is determined by the audited results, not by the number of model repositories archived.

3 A finite-support decision audit

The object being audited. An audit context x fixes a physical root state, task goal, executable candidate bank $\mathcal{A}_x = \{a_1, \ldots, a_K\}$, horizon, continuation policy, observation encoding, and utility definition. A candidate may be an action chunk; its continuation must be the same under every comparison. Let $u_k = U(x, a_k)$ be its utility measured by environment execution. Higher utility is better, so a cost is negated before applying the definitions. Success, return, terminal reward, and time-to-completion are distinct utilities and receive separate audits. Initially, u_k denotes a fixed cached outcome; the extension to expected outcomes is given below.

The model supplies forecasts $\widehat{z} = (\widehat{z}_1, \ldots, \widehat{z}_K)$. A frozen scoring and selection rule σ produces a distribution $p = \sigma(\widehat{z})$ on the bank; deterministic selection is a point mass. We write $U(p) = \sum_k p_k u_k$. The audited intervention may replace the forecast supplied to this selector. It must not change the task goal, candidate actions, root state, scoring parameters, continuation policy, utility implementation, or the coupling of random streams. A system whose native policy does not consume forecasts can be studied only through a separately identified, added planner. Neither kind of audit establishes an adversarial attack without a specified feasible intervention and observed utility loss.

3.1 Leverage and baseline-relative harmful headroom

Define

$$u_{\max} = \max_k u_k, \qquad u_{\min} = \min_k u_k,$$
$$\mathrm{Lev}(x) = u_{\max} - u_{\min}, \qquad H_b(x) = U(p_b) - u_{\min}, \tag{1}$$

where p_b is the baseline selector. The first quantity is the width of the feasible utility interval. The second is the maximum loss relative to this baseline when unrestricted selection within the recorded bank is allowed; a constrained intervention can attain less.

Proposition 1 (finite-support bound). For any two distributions p, q on the same bank,

$$|U(p) - U(q)| \le \mathrm{Lev}(x)\,\mathrm{TV}(p,q) \le \mathrm{Lev}(x), \qquad U(p_b) - U(q) \le H_b(x), \tag{2}$$

where $\mathrm{TV}(p,q) = \frac{1}{2}\sum_k |p_k - q_k|$. For a panel with nonnegative weights w_x summing to one, the corresponding aggregate upper bounds are the weighted sums of these bounds.

Proof. Subtract $u_{\min}$ from every utility; the difference of expectations is unchanged because both distributions sum to one. On coordinates with $p_k > q_k$, the positive contribution is at most $(u_{\max} - u_{\min}) \sum_{p_k > q_k} (p_k - q_k)$; negative contributions cannot increase it. This sum is $\mathrm{TV}(p,q)$. Interchanging p and q gives the absolute bound. Every expectation is at least $u_{\min}$, which gives the baseline bound. Nonnegative weighted summation preserves the inequalities. Selecting an extreme-utility candidate attains the interval endpoints when unrestricted selection is permitted.□

These inequalities are elementary consequences of finite support, not a new robustness theorem. Their use is to check whether an interface can exhibit a practically meaningful effect before spending model calls. They require real branch outcomes, and are therefore not free to compute when the cache does not yet exist. A deterministic top-choice flip is necessary for a utility change on this bank, but a flip between equal-utility candidates has zero effect. Changes in lower ranking positions are weaker evidence still. A bound violation calls for checking candidate coverage, continuations, replay randomness, and utility definitions; it does not refute the finite-support identity. A small bound on one bank is not a guarantee about unrecorded actions, later replanning, or other states.

Averaging actions is not randomizing over actions. If a planner executes an elite mean $\bar{a} = \sum_k w_k a_k$, its utility is $U(x, \bar{a})$, not $\sum_k w_k U(x, a_k)$. For example, the constructed utility $U(a) = a^2$ gives utility one for both recorded actions -1 and 1, but zero for their mean. Thus their recorded leverage is zero while the unrecorded mean loses one unit. The actual mean must be replayed and included in the support; changes in an adaptive CEM search distribution are also outside a frozen-bank bound unless their outputs are covered.

3.2 A signed true-future comparison

Let $z^\star$ be the encoding of the real future for each recorded candidate, at the same prediction horizon and with identical preprocessing. Keep the scorer and tie-breaking rule fixed and define $p^\star = \sigma(z^\star)$. Then

$$\begin{aligned} \mathrm{Reg}(x) &= u_{\max} - U(p), \\ \mathrm{Mis}(x) &= u_{\max} - U(p^\star), \\ \mathrm{Pred}(x) &= U(p^\star) - U(p), \\ \mathrm{Reg}(x) &= \mathrm{Mis}(x) + \mathrm{Pred}(x). \end{aligned} \tag{3}$$

The identity follows by adding and subtracting $U(p^\star)$. Mis ≥ 0 is the residual regret of the frozen representation–scorer–selector interface when supplied encoded true futures. It need not be solely a scorer-weight defect: lossy representations, horizon mismatch, utility-proxy mismatch, ties, and an approximate selection rule can all contribute. Encoded true futures do not provide a perfect physical state representation or the optimal full-horizon planner.

In contrast, Pred is signed. A negative value means that model forecasts select a higher-utility candidate than true-future substitution does. It can reflect compensating errors. The two terms must not be presented as nonnegative shares of an error budget. The algebra alone also does not identify causal mediation through a complete native planner: it is a controlled replacement comparison at a declared frozen interface. True futures are an offline diagnostic and cannot be used as an available deployment input or as labels for a test-set-trained defense.

Two constructed examples clarify the claims. Take two actions with utilities $(1, 0)$ and a selector maximizing $-z_k$. If the true features are $(0, 1)$ but the model predicts $(1, 0)$, then Mis $= 0$ and Pred $= 1$: prediction error causes regret despite a correctly ordered true-future proxy. Objective mismatch is therefore not a necessary condition for predictive error to harm decisions. Conversely, if the true features are $(1, 0)$ and the model predicts $(0, 1)$, then Mis $= 1$, Pred $= -1$, and Reg $= 0$. Replacing the forecast with the truth reduces its squared prediction error to zero but decreases utility. These are mathematical counterexamples, not experimental results from any audited model.

Forecast error is different from goal cost. Forecast error compares $\widehat{z}_k$ with the real consequence $z_k^\star$ of the *same* action. Distance between an encoded outcome and a desired goal is a planning cost, even when both quantities are called visual MSE. A lower goal distance together with lower task return demonstrates proxy disagreement; it does not establish improved predictive fidelity. In particular, Mis > 0 alone does not prove the hypothesis that improving forecasts makes decisions worse.

3.3 Uncertainty and identifiable follow-up tests

For a stochastic environment, replace u_k by $\mu_k = \mathbb{E}[U(x, a_k)]$ to obtain a population statement for the *fixed* bank and continuation. A single replay only supplies a realized outcome; its empirical extrema do not bound these expectations. If simultaneous intervals $[\ell_k, r_k]$ cover all μ_k with probability at least $1 - \alpha$, then on this event

$$\mathrm{Lev}_\mu \leq \max_k r_k - \min_k \ell_k, \qquad H_{b,\mu} \leq \sum_k p_{b,k} r_k - \min_k \ell_k. \tag{4}$$

For bounded returns in $[A, B]$ and n_k independent replications per candidate, one elementary construction uses sample means plus or minus $(B - A)\sqrt{\log(2K/\alpha)/(2n_k)}$, clipped to $[A, B]$. This follows from Hoeffding's inequality [7] and a union bound. The within-arm independence assumption is necessary; shared random numbers across arms do not themselves invalidate the union bound. Reusing one realized future both as an oracle selector input and its evaluation can yield a clairvoyant choice; it is not the expected performance of an implementable policy. Scoring an expected latent is also generally different from averaging scores of stochastic futures.

A fidelity–utility reversal test must compare two specified forecast inputs on identical candidates, show an improvement under a preregistered forecast-error metric, and measure a decrease in independently evaluated selected-action utility. An offline interpolation toward true futures can test that interface property, but cannot claim a deployable accuracy improvement. Establishing

positive mean Mis is a separate test of residual regret. Establishing that scorer repair outperforms forecast repair requires a direct, paired comparison of held-out utilities under both interventions, together with their unmodified controls. The identity does not predict which feasible repair will win. Calibration roots, model or metric selection, and final evaluation roots must be separated. Independent episode or initial-state clusters, rather than candidates or multiple roots from the same episode, determine the sampling unit. Thirty roots is a possible design size, not a guarantee of power; uncertainty and minimum detectable effect depend on variability and the sampling design.

4 Evidence, systems, and evaluation protocol

4.1 Evidence provenance

This study combines archival reanalysis with a completed prospective experiment. The Cosmos panel aggregation and development objective-weight audit reuse saved model outputs and continuations without new inference or environment stepping. The DINO confirmation panel instead executes 150 candidate continuations at 30 new roots: 4500 candidate controls, 30 forecast unrolls, and 240 encoding calls. Its recorded CPU invocation takes 897.12 seconds, loads the model once, and makes no model API calls, network attempts, training updates, or CUDA calls. The separate expert-goal setups and root warmups add 3000 and 30 explicit controls; constructor-internal simulation is not instrumented. Subsequent aggregation uses cached outputs. Four later diagnostics are distinguished from this confirmation: R003 executes one native CEM query and 165 candidate prefix controls; R004 phase 1 fits and evaluates a fixed bias entirely from caches; and R004 phase 2 executes two native CEM queries and 60 replay controls at a separate root. R004 phase 3 runs complete native episodes at that same root, with 14 planning queries and 198 policy controls across the two arms. Its worker completes in 343.45 seconds and records a sampled process peak of 5710 MiB. Each native diagnostic has an expert-goal setup and warmup, with phase 3 reproducing phase 2's root and Goal exactly. These exploratory follow-ups are not additional confirmation roots for the original oracle test. R005 separately crosses saved TD-MPC2 dynamics modules between their native proposal and evaluation roles, executing four complete 500-control episodes in 90.68 seconds without training or API calls. This is one reused development case, not four independent roots or a new attack. Sources and recomputation entry points appear in Appendix A. We retain technical failures, stopped gates, and protocol corrections as such, rather than counting them as failed tasks or silently replacing their results.

Only measurements with an identified root, fixed candidate identity, declared continuation, and independently measured physical utility can enter a utility support table. Model values, latent distances, and predicted rewards cannot serve as the ground truth. A table may be complete for a restricted interface without being complete for the planner's action space.

4.2 Cosmos Policy: a fixed first-choice interface

The audited implementation uses a candidate–future–value path: generate an action candidate, predict a future conditional on it, and score that future before selecting a candidate. The development panel contains ten LIBERO-Goal tasks with three initial states per task. For each root, both saved first candidates have real continuations. The selected first block contains 16 physical controls; subsequent planning uses the original task Goal and the recorded native continuation settings. The horizon cap is 300 controls.

We evaluate success and capped completion steps separately. Cost is converted to utility by sign reversal where needed in the formal definitions. A success tie does not imply a cost tie.

The panel has a deterministic recorded outcome for each branch; it is not a confidence bound on repeated stochastic rollouts. Tasks are the natural grouping variable for any exploratory uncertainty calculation. Thirty roots and sixty branches do not supply sixty independent replicates.

The Goal-Swap diagnostic changes only the Future branch's text, while the Policy and Value branches retain the original task. We distinguish a cache-derived first-choice contrast from a separately executed continuous-Swap episode. A join that selects an already measured continuation is an offline evaluation, not a newly executed recovery episode. The local Torch compatibility adapter also remains part of the evaluated implementation; these measurements do not establish parity with every upstream numerical backend. This is a researcher-controlled internal Future-context intervention. A legal alternative Goal in the same scene does not establish an externally writable interface, joint training-support membership, or a universal Goal-invariance obligation for observational predictions.

4.3 DINO-WM reproduction: fixed action banks

Our checkpoint and planner come from the JEPA-WMs project's DINO-WM reproduction [13]. The native planner uses CEM with 300 samples, 15 iterations, 10 elites, six macro-actions per plan, and five physical controls per macro-action. It normally executes three macros before replanning. The task Goal is used in the scoring objective; it is not a separate Goal field in this action-conditioned predictor. Thus the operation here is not Goal-Swap.

The historical two-action diagnostic fixes the first complete mean returned by each of two native planning conditions as a_0 and a_1. These means were selected post hoc and replayed from the same root for all 30 controls. The first 15 controls match the saved native prefixes. The remaining controls are an open-loop suffix, not the replanning behavior that the native controller would have executed. They form a newly evaluated two-action bank, not the native 300-sample population. In particular, a CEM elite mean is a new executable action sequence, not a random choice of an elite trajectory.

Within each action, canonicalization replaces saturated XYZ request values with boundary representations: after FP32 action denormalization, clipping to $[-1, 1]$ gives exactly the same XYZ controls. Unsaturated coordinates and the raw gripper command are unchanged. This preserves execution, but does not establish membership in the predictor's training support. The root, original Goal, candidate actions, and scorer are fixed in the bank comparison. Real endpoint re-encoding, with exact parity checks against saved encodings, supplies the reference for feature forecast error and for oracle-feature scoring. This measures feature fidelity, not complete physical-state prediction accuracy.

The unchanged native terminal objective is

$$C(z, z_g) = \mathrm{MSE}(z^v, z_g^v) + 0.1\,\mathrm{MSE}(z^p, z_g^p). \tag{5}$$

Here v and p denote visual and learned proprioceptive features. The Goal is an achieved expert-example observation. The proprioceptive features encode hand position and finger separation; their squared distance is not the task's TCP-to-target distance. We distinguish $C(z^\star, z_g)$, a real-future-to-Goal *proxy cost*, from $\mathrm{MSE}(\hat{z}, z^\star)$, a *forecast error*. Confusing these quantities would invalidate the prediction-improvement claim.

4.4 Prospective DINO oracle-substitution protocol

The confirmation experiment uses the frozen five-action bank from development: 30-control sequences originating from seeds 1, 2, and 3, a physical-zero sequence (candidate3), and an XYZ

sign-reversal variant of the seed1 sequence (candidate4). All five sequences are transferred unchanged to every test root. The selector is an added argmin ranker using the reproduction's unchanged encoder, predictor, and Equation 5; it is not a native adaptive CEM run. Each candidate is committed open loop for 30 controls.

The 30 confirmation roots use independently seeded fresh MetaWorld reach resets, seeds 1001–1030, after the native one-control warmup. Development seeds 1–4 and engineering-calibration seed 1000 are excluded. Each root has its own unchanged expert-observation Goal. Restoration checks match the root's rendered pixels, proprioception, and integration state before each candidate replay. All scheduled roots are complete and retained, including ties and low-utility outcomes. These seeds were unused by this project; they are not established as outside the model's training data.

One original forecast supplies all five candidate predictions at a root. Actual observations at controls 5, 10, 15, 20, 25, and 30 are independently replayed and encoded. After candidate actions and the original forecast are fixed, diagnostic arms replace future features by $z_\lambda = (1-\lambda)\hat{z} + \lambda z^\star$ for the prespecified grid $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$. The endpoints use the exact cached prediction and actual encoding, respectively. The context, original Goal, candidate identities, physical controls, objective coefficient, and reduction batch shape are held fixed; exact scalar cost ties select the smallest candidate index. Intermediate mixtures may lie outside the learned feature manifold. Actual-future information is an offline scoring instrument, not information available to a deployed controller.

The primary utility is terminal environment reward after 30 controls, aligned with the scorer's terminal objective. Forecast fidelity is the mean across the five candidates of terminal visual-feature MSE plus 0.1 times proprioceptive-feature MSE against each candidate's actual encoding, evaluated in FP64; scorer costs retain CPU FP32 arithmetic. The primary contrast is oracle minus original selected utility, paired within each root. The preset criterion requires a negative one-sided 95% bootstrap upper bound on its mean and strictly decreased forecast error. Oracle error is zero by construction, so the latter is a manipulation check, not an independent discovery of improved prediction. Positive oracle mismatch alone does not satisfy this criterion.

We use 10,000 whole-root paired bootstrap resamples with NumPy `default_rng` seed 2026101601 and report two-sided percentile 95% intervals alongside the primary one-sided bound. All actions and arms remain paired within each sampled root; the sample size is 30, not 150 continuations or 750 root–candidate–arm combinations. Root event rates use Wilson 95% intervals. Cumulative reward, distance, success, the intermediate mixture grid, and 15-control prefix utilities are descriptive secondary endpoints. Every arm selects using the 30-control score; prefix outcomes do not constitute a separate 15-control oracle selector. Fixed candidate choices, physical zero, uniform-choice expectation, and per-root best-utility hindsight references are reported without selecting a new baseline on the confirmation outcomes.

4.5 Native averaging and a fixed residual-correction follow-up

The R003 check executes unchanged native CEM at fresh physical seed 3000 with the settings above and local planning seed 1. It records the final population, all ten elites, and the returned elite mean, verifying that the executed prefix matches that mean exactly. Each elite and the mean is replayed from the same root for the native 15-control execution prefix. The preset condition is that the mean's terminal reward is below every elite's terminal reward by more than 10^{-6}. This is a diagnostic of an executable mean's utility, with no external adversary or forecast intervention.

R004 phase 1 addresses the availability limitation of oracle substitution. For each feature block

and each future horizon t, it fits the vector

$$b_t = \frac{1}{5}\sum_{k=1}^{5}(z^{\star}_{t,k} - \widehat{z}_{t,k}), \qquad \widetilde{z}_{t,k} = \widehat{z}_{t,k} + b_t, \tag{6}$$

using only the five original-forecast residuals at development seed 1. The bias is independent of the candidate at application time and varies by horizon and feature coordinate. It is saved before evaluation data are read, then reloaded without refitting, scaling, utility-based selection, or use of evaluation futures. Original context features remain unchanged. Application requires only a new forecast and the stored bias, but the term "calibration" in artifact names does not imply probabilistic calibration, coverage, or a statistical calibration guarantee. This is an additive residual correction.

Phase 1 reuses all 30 previously examined roots and the same five-action bank; it is an exploratory evaluation, not a new independent confirmation. The fixed-bias primary error averages forecast-to-actual-future weighted MSE over six future horizons and five candidates, excluding context; terminal forecast MSE is reported separately. Neither is distance to the task Goal. The scorer still minimizes terminal Goal distance using the native FP32 formula with batch size five. The original costs and choices are checked against the cached originals. All roots, candidate baselines, and outcome ties are retained.

R004 phase 2 applies this same saved correction, without refitting, inside two full native CEM queries at fresh root 3001, one original and one corrected. Both use 300 samples, 15 iterations, ten elites, local seed 1, and no microbatching. Their initial populations match; later proposal populations change endogenously with corrected scoring. This is an adapted native CEM comparison, not a fixed-population argmin experiment. The actual returned means are committed before real replay and are each executed for 30 controls, with the first 15 reported as the native execution prefix. Controls 16–30 continue the saved mean instead of replanning. A separate common batch of the two returned means measures original and corrected forecast MSE against their real six-milestone futures; it does not measure fidelity for every CEM proposal.

The executable phase-2 gate specifies lower mean six-horizon forecast MSE on these two means and lower H15 terminal reward. A requirement for lower H15 cumulative return was added after executable launch but before target replay outcomes were observed; we report that timing rather than presenting the expanded criterion as fully prelaunch. Results at both H15 and H30 are retained. The correction is an explicitly added adapter. A third-party forecast-provider permission is a proposed service integration, not a verified external write surface of the native implementation or an established attack.

R004 phase 3 tests whether the prefix loss persists over a complete episode. It deliberately reuses inspected root 3001 and the same Goal and saved bias; both arms' first query must exactly reproduce their phase-2 actions, full mean, and initial population. The environment runs to its native 100-step limit: one warmup control followed by 99 policy controls, with `done_at_succ=False`. Each arm makes seven native planning calls, using the latest native wrapped observation, a continuous per-arm RNG stream, and the native per-query initialization rather than adding a warm start or history mechanism. The last query naturally has a two-macro planning horizon; the correction uses the corresponding prefix of the saved horizon-specific bias and preserves the actual context. There is no refit or seed search. The preset primary condition is corrected minus original complete-episode cumulative return below -10^{-6}. Final success, ever-success, terminal reward and distance, and 15-control cumulative-return checkpoints are secondary. This protocol measures complete-episode utility; it does not measure forecast accuracy across the complete episode.

Measured quantity	Support minimum	Native	Support maximum
Successes, out of 30	28	29	30
Total capped completion steps	3230	3477	3646

Table 1: Cosmos finite-cache bounds. The success and step extrema optimize different metrics and need not be attained by the same selector. Per-root extrema use hindsight outcomes; they do not demonstrate a realizable attack.

5 Results

5.1 Finite support sharply limits the Cosmos success contrast

For the 30-root Cosmos table, the sum of per-root minimum successes is 28 and the sum of maxima is 30. The recorded native selector attains 29. Therefore its maximum additional failure count under a first-choice-only substitution is one, or 3.33 percentage points over this panel. The total support width is two successes; this is different from the baseline-specific harmful headroom. Twenty-eight roots have identical binary outcomes for both candidates.

There is substantially more variation in capped completion cost: 20 roots have unequal costs. Any selection over the recorded support has total capped steps between 3230 and 3646; the native total is 3477. Its maximum possible increase is therefore 169 steps, or 4.86%. This bound says nothing about later interventions, a changed candidate generator, another continuation, or unobserved roots.

Fixed candidate0, fixed candidate1, uniform choice in expectation, and the native first selector each yield 29 successes on this table. The native ranking thus has no aggregate binary advantage over these simple selectors on the observed panel. Nevertheless, it matters locally. At task3/init2, candidate1 succeeds in 170 controls and candidate0 fails at the 300-control cap. The recorded Swap first choice selects candidate0. Its cache-derived first-choice evaluation yields 28 successes and 130 additional capped steps across the panel. This already reaches the binary floor relative to the recorded native selector; searching a different first-choice rule cannot produce three new failures on these same branches.

This is a useful screening conclusion before additional model calls, but not before obtaining the real continuation table. Computing the bound is cheap only because the required branch executions already exist. It does not explain every failed intervention in the research archive.

5.2 Oracle substitution lowers utility on 30 independent DINO roots

The completed panel satisfies the prespecified offline fixed-bank criterion. Mean terminal reward decreases from 6.300613 under original predictions to 4.758220 under actual-future scoring (Table 2). The paired mean change is -1.542393, with two-sided 95% bootstrap interval $[-2.438788, -0.715561]$ and one-sided 95% upper bound -0.842993. Mean terminal feature forecast MSE changes from 0.974129 to exactly zero. Its reduction at every root is guaranteed by oracle substitution and does not demonstrate a learned or deployable forecast correction.

Terminal reward decreases at 11 of 30 roots (36.7%; Wilson interval 21.9–54.5%), remains equal at 18, and improves at one. The selected candidate changes at 12 roots (40.0%; Wilson interval 24.6–57.7%). Thus the negative mean is not a decrease at every root; the paired median change is zero. Mean cumulative 30-control return also decreases, from 109.127186 to 90.311600, and mean terminal TCP-to-target distance increases from 0.123928 to 0.161616. Terminal success and ever-success both change from 2/30 to 0/30. These success counts are descriptive secondary outcomes;

Selector	Mean terminal reward	Mean 30-control return
Original forecast ranker	6.300613	109.127186
Actual-future oracle ranker	4.758220	90.311600
Always candidate0	7.306371	115.748872
Always candidate1	4.255455	83.645909
Always candidate2	7.262187	119.205578
Physical zero (candidate3)	1.482745	44.415023
Always candidate4	0.485454	25.037217
Uniform choice (exact expectation)	4.158442	77.610520
Per-root best utility (hindsight)	7.992185	128.161229

Table 2: Completed prospective DINO panel: 30 independently reset roots, the same five development action sequences per root, and 30 open-loop controls. All selectors use the same real candidate replays. The two hindsight columns optimize their respective utility separately and need not select the same candidate. Hindsight is an unavailable upper reference, not a deployable scorer. The original and oracle rankers both have lower mean utility than fixed candidate0 and candidate2 on both reported metrics.

they do not establish a separate binary-success claim.

The simple comparators materially constrain interpretation. Always choosing candidate0 or candidate2 achieves mean terminal reward 7.306371 or 7.262187, respectively, exceeding the original ranker's 6.300613. Both rankers exceed uniform choice in expectation and physical zero, but the original ranker shows no aggregate advantage over these two fixed sequences. Identifying candidate0 as the best fixed choice for terminal reward from these test outcomes is a post-hoc oracle comparison; it is not a scorer fitted on separate training roots. The experiment therefore confirms an oracle-substitution effect in a weak fixed-bank selector, without establishing normal native planner competence or a validated scorer repair.

For terminal reward, the mean support range is 7.506731, mean oracle-feature mismatch is 3.233965, and the original ranker's signed prediction term is -1.542393. Its mean regret is consequently $3.233965 - 1.542393 = 1.691572$ up to rounding. This is an algebraic account of the observed choices: prediction error partly offsets this proxy's mismatch on average, without identifying causal shares of a native planner's failure.

The preset intermediate mixture arms have mean terminal rewards 5.898811, 5.718824, and 5.227515 at $\lambda = 0.25, 0.5, 0.75$, with corresponding forecast MSEs 0.547948, 0.243532, and 0.060883. The observed mean-utility sequence decreases as oracle information increases; this descriptive grid neither establishes rootwise monotonicity nor evaluates an independently improved predictor. For the selected actions' first 15 controls, mean terminal reward changes from 3.293546 to 2.856326 and mean cumulative return from 32.592677 to 30.330693. These are prefixes of the same 30-control selections, not native replanning evaluations.

5.3 The native elite-mean gate is negative

R003 completes in 56.18 seconds and does not meet its preset condition. The native mean's H15 terminal reward is 3.690524, inside the ten-elite range $[3.538413, 3.887301]$ and 0.152111 above its minimum. Four elites have higher terminal reward than the mean, but this does not establish the specified below-all-elites effect. The mean's cumulative return is 31.635962; none of the eleven branches succeeds within 15 controls. The gate is closed without a seed sweep. This negative observation neither invalidates the finite-support distinction between randomizing and averaging

Execution	Utility	Original	Fixed bias	Difference
H15 native prefix	Terminal reward	3.013245	2.730845	−0.282399
	Cumulative return	29.943025	28.036593	−1.906432
H30 open-loop mean	Terminal reward	5.231756	7.399076	+2.167321
	Cumulative return	98.235031	100.029401	+1.794370
H30 native replanning	Terminal reward	7.381968	8.070076	+0.688107
	Cumulative return	112.651169	107.150320	−5.500850
H99 native replanning	Terminal reward	9.996284	10.000000	+0.003716
	Cumulative return	747.432862	760.171563	+12.738701

Table 3: R004 phases 2 and 3 at the same root: execution horizon and continuation change the comparison. H15 is the shared initial native prefix. The H30 open-loop row continues the first committed mean; H30 native replanning is the first 30 controls of the complete 99-control episode. Only H99 measures the complete policy episode, excluding its warmup. Differences are fixed bias minus original. The repeated root does not supply independent replications.

actions nor establishes a general guarantee for CEM means.

5.4 A fixed residual correction reproduces a bounded exploratory effect

On the reused 30-root bank, the frozen development-fitted correction reduces mean six-horizon forecast MSE from 0.777127 to 0.511728 and terminal forecast MSE from 0.974129 to 0.636567. Mean selected terminal reward decreases from 6.300613 to 5.895755, and cumulative H30 return from 109.127186 to 103.668100. Reward decreases at four roots, ties at 25, and increases at one; five selected actions change. Terminal and ever-success both change from 2/30 to 0/30, descriptively. All 30 roots have lower candidate-averaged error, but this does not mean every candidate improves: physical-zero candidate3 has higher mean and terminal forecast error at every root. Both rankers still trail the fixed candidate0 and candidate2 baselines in Table 2. No new population inference is made from this adaptively reused panel.

The separate native CEM query shows a more limited, horizon-dependent result (Table 3). On the common two-mean diagnostic batch, mean six-horizon forecast MSE falls from 1.361239 to 0.978856 and terminal forecast MSE from 1.927436 to 1.569059. The corrected query's actual H15 terminal reward is lower by 0.282399 and its H15 cumulative return by 1.906432, meeting both the executable gate and the pre-outcome expanded criterion. At H30, however, terminal reward is higher by 2.167321 and cumulative return by 1.794370. Neither branch succeeds at either horizon. This is one root and two distinct adaptively generated mean actions, not an independent population replication or a claim that accuracy improved throughout the CEM search.

Complete-episode return loss is absent. Both phase-3 arms complete 99 policy controls with native replanning. Cumulative return is 747.432862 for the original and 760.171563 for the fixed bias, a change of +12.738701. The preset persistent-return-loss condition is false, and this fixed gate is closed without refitting or a seed sweep. The H30 native-replanning prefix still has lower cumulative return with the correction, 112.651169 versus 107.150320; by the H45 checkpoint the direction reverses, 225.690382 versus 248.581864, and remains positive at subsequent reported checkpoints. These H30 values are distinct from the phase-2 open-loop returns of 98.235031 and 100.029401.

Final success changes from false to true and final target distance from 0.052013 to 0.046815. Both arms achieve success at some time, but first success occurs later with the correction: policy

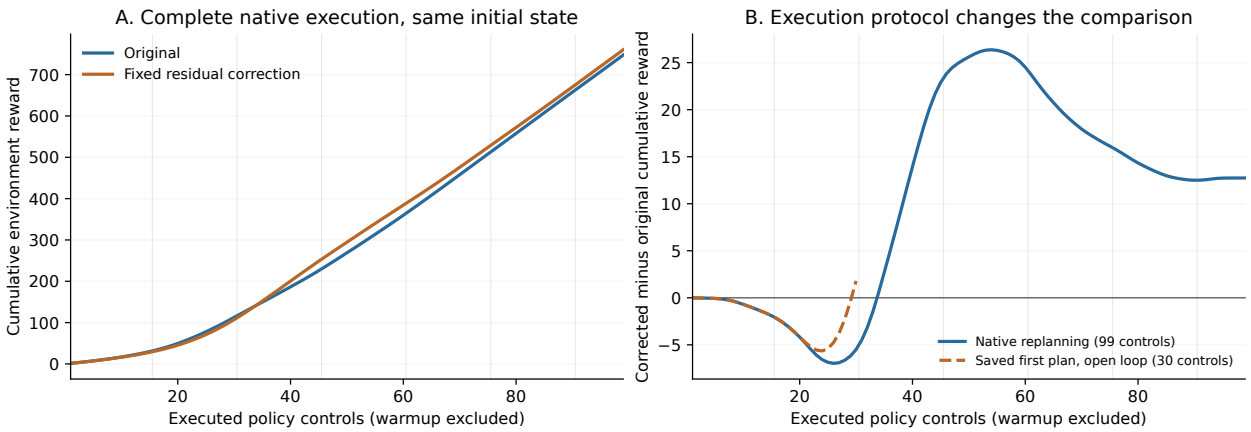


Figure 1: R004 at the same inspected root. A: complete native replanning episodes, with 99 policy controls after one warmup control. B: fixed-bias minus original cumulative return under native replanning (solid) and continuation of the saved first mean for 30 open-loop controls (dashed). The shared initial prefix does not make the subsequent continuations interchangeable. Complete episode return improves, so the persistent-loss gate is negative. This is an exploratory single-root comparison; initial-query forecast-error improvement does not establish full-episode forecast accuracy.

control 97 versus 72. The delayed first success is a genuine contrary secondary observation, not a substitute for the prespecified cumulative-return endpoint. Because `done_at_succ=False`, both arms continue after success to the same environment step limit. Figure 1 displays the cumulative returns and distinguishes the two continuation protocols.

The follow-up therefore supplies a deployable-form feature correction that improves the declared initial-query forecast metric while reducing utility in a fixed bank and one execution prefix, followed by a complete-episode result in the opposite direction. It supplies neither a calibration guarantee nor evidence of persistent return degradation. Phase 3 reuses the same inspected root; its implementation checks are not additional independent evidence of generality, and its complete-episode forecast accuracy has not been measured.

5.5 A development canonicalization improves fidelity but lowers return

Table 4 reports the fixed two-mean diagnostic. The original prediction scores select a_0; canonical-request scores select a_1. For both actions, canonicalization reduces weighted terminal feature forecast MSE. Yet the independently replayed 30-control return decreases by 19.8048, and terminal environment reward decreases by 1.9009. Neither branch succeeds within these 30 controls, so this comparison shows no binary success loss. The selection is made using the Goal proxy, not the forecast-error metric: the predicted cost gap $C(a_1) - C(a_0)$ changes from $+0.265703$ to -0.344738, while the actual-future proxy gap is -0.321822.

The actual endpoint features also select a_1. With cumulative return as utility, the bank optimum is $U(a_0) = 95.841823$ and oracle-feature utility is $U(a_1) = 76.036994$. Objective mismatch regret is 19.804829. The original selector has zero total regret, decomposed as $19.804829 + (-19.804829)$; the canonical selector has regret 19.804829, decomposed as $19.804829 + 0$. Prediction error in the original condition happens to offset the scorer's mismatch on this bank. The negative prediction term is essential and must not be clipped to zero or reported as a causal percentage.

This development effect is not a population estimate. The root and actions were involved in

Action	Original forecast MSE	Canonical forecast MSE	True-future proxy cost	Real return	Terminal reward
a_0: original returned mean	1.565623	0.529210	1.701293	95.841823	5.247907
a_1: canonical returned mean	0.958529	0.458303	1.379470	76.036994	3.347029

Table 4: One post-hoc root, two fixed actions, 30 physical controls. Forecast MSE is measured against each action's real endpoint. True-future proxy cost is Equation 5 evaluated against the original Goal; it is not forecast error. Real return and terminal reward come from the environment. Both actions have zero successful controls.

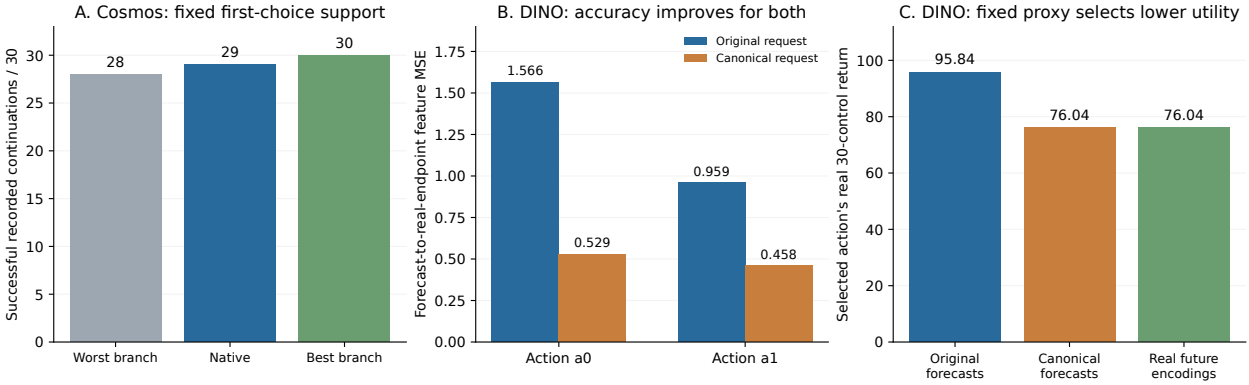


Figure 2: Two distinct decision diagnostics. A: fixed-support success limits on Cosmos. B–C: more accurate forecasts and lower selected real return coexist at one DINO root. The oracle-feature selector also chooses the lower-return action. Panels B and C share the same two actions and are not independent experiments. No population uncertainty is estimated from this single root.

mechanism development, and forecasts would change candidate generation inside a full CEM rerun. Consequently, these measurements cannot identify the behavior of a native planner equipped with an exact world model. The companion five-action bank comes from development seed1, whereas this two-mean bank comes from seed4. They are two distinct physical roots, but use different candidate-construction protocols and were examined adaptively. They cannot be pooled as a preregistered confirmation sample. On the seed1 bank, canonicalization lowers selected terminal reward from 7.699376 to 1.829723, and actual-future scoring selects the same lower-reward action. The highest-reward action in that bank attains 7.989219. All five actions fail the binary success criterion over the 30-control horizon.

5.6 The proposed component-weight repair fails an offline necessity check

The small native proprioceptive coefficient might suggest that increasing it would fix the DINO mismatch. The measured actual-future cost differences contradict that proposed explanation:

$$C_v(a_1) - C_v(a_0) = -0.321716905, \tag{7}$$

$$C_p(a_1) - C_p(a_0) = -0.001053669. \tag{8}$$

For any $w_v, w_p \geq 0$ with at least one positive, $w_v \Delta C_v + w_p \Delta C_p < 0$ in real arithmetic. Thus every nonnegative, nonzero weighted combination still prefers the lower-return action in this oracle-feature bank. An illustrative seven-weight grid reproduces this result; the argument covers the entire family

rather than choosing a successful weight after inspecting test outcomes. All-zero weights give an uninformative tie, not a calibrated scorer.

This rules out one inexpensive repair *for the observed oracle-feature ordering*. It does not rule out task-specific features, a different reward model, nonlinear scoring, or effects on model-predicted features. It also does not establish that scorer-side interventions outperform forecast-side ones. Such a comparison requires separate development data, matched resource budgets, and held-out real outcomes.

5.7 Cosmos also contains a local prediction-side failure

Objective mismatch is not required for every prediction-induced decision error. At the already identified task3/init2 root, original future features select the successful candidate, Swap features select the failed candidate, and real-future feature substitution selects the successful one. Swap visual latent MSE is 10.37 and 9.50 times its original value for the two candidates. This is a local example in which larger forecast error, worse selection, and oracle-feature recovery align.

A saved 3×3 factorial uses original, Swap, and real sources for visual and proprioceptive future slots. Under each of the three proprioceptive backgrounds, Swap visual slots select the failed branch, while original and real visual slots select the successful branch. This identifies conditional dependence of the frozen Value input on visual provenance at this root. The nine cells share two historical continuations; they are not nine tasks, new recovered episodes, or a deployable defense. Mixed feature blocks need not describe a physically coherent future.

5.8 TD-MPC2: separating proposal and scoring consumers

We test a different explanation for an existing development-case return gap: can a dynamics update cause harm primarily by changing the policy proposals, so that restoring the scoring path cannot recover performance? We reuse two saved dynamics modules fitted from the random-source (R) and cascade-source (C) `cheetah-run` trajectories at source seed 23. R is an adapted comparison model, neither the original pretrained Clean model nor an oracle for physical outcomes. No new fitting is performed.

The native TD-MPC2 planner uses dynamics both to generate 24 policy-prior trajectories and to evaluate its 512 candidates. We independently assign R or C to these two consumers, preserving the native horizon 3, six optimization iterations, 64 elites, stochastic action selection, normalization, and all other original model weights. The scoring route includes imagined-state reward readouts and terminal policy/Q evaluation; it is not a direct forecast-distance score. Each arm executes one complete 500-control `walker-run` episode from seed 31, with the same initial state and reset random streams. Later states, proposal populations, and optimizer warm starts may diverge as endogenous consequences of the intervention.

Arm	Proposal dynamics	Scoring dynamics	Actual episode return
RR	R	R	612.082555
CC	C	C	495.216321
CR	C	R	674.885888
RC	R	C	553.538964

Table 5: One reused TD-MPC2 development case. Returns are sums of actual 500-control environment rewards, not planner value estimates.

Both RR and CC exactly reproduce the saved observations, actions, per-control rewards, and complete returns. At the common first query, all arms share the 24 prior first actions and all 488 Gaussian candidate trajectories exactly; prior continuation actions can differ. Recording these already-generated candidates preserves both CPU and CUDA random states and adds no model evaluation.

The preregistered condition fails. Writing $G = J_{RR} - J_{CC} = 116.866234$, it requires both $J_{RR} - J_{CR} \geq G/2$ and $J_{RC} - J_{CC} \geq G/2$. Instead, J_{CR} exceeds J_{RR} by 62.803333, while the second contrast recovers 49.9055% of G. In particular, retaining C proposals and restoring only R scoring raises return from 495.216321 to 674.885888. The factorial interaction $J_{CC} - J_{CR} - J_{RC} + J_{RR} = -121.125975$ also precludes an additive allocation of the original loss to the two routes. This counterexample does not support the proposed claim that damaged proposal coverage prevents scoring-path recovery in this case. It does not measure the physical quality of every candidate, demonstrate improved forecast accuracy, or establish an attack, a defense, or a population effect. The fixed gate is closed without a seed or threshold sweep.

SWAAP already describes poisoned world-model adaptation and the use of TD-MPC2 dynamics for both policy-guided proposals and candidate evaluation [8]. Our distinction is the inference-time factorial separation of these consumers, not a first demonstration of model poisoning or proposal dependence. Here that separation contradicts the registered proposal-dominant explanation. Results and the launch receipt are in `runs/tdmpc_proposal_scoring_factorial_v1/summary.json` and `runs/tdmpc_proposal_scoring_factorial_v1/launcher.json`; all four physical traces and the common-first-population captures are retained.

5.9 A complete-person ordering control in GenSafeNav

R006 tests an explicitly added representation boundary at a learned forecast consumer. After native prediction and feedback processing, we reverse the visible prefix of person rows presented to the pretrained GenSafeNav actor. Each person's current position, all forecast coordinates, and conformity features move together; padding, visibility count, robot inputs, physical entities, and all weights are unchanged. Inverse-permutation checks verify that no feature values or associations are lost. This differs from assigning one person's prediction to another person.

We reuse the existing case-0, seed-42 CPU protocol, including zero conformity/cost noise, collision penalty -20, deterministic actor actions, the CPU device bridge, and the existing native feedback indexing behavior. Each arm retains its own normal aggregated recurrent state. The episodes share exactly the same initial physical snapshot; later states and predictions may diverge as consequences of different controls. No forecast-equivariance claim is tested by the actor-only intervention.

Both complete episodes reach the goal after 119 physical controls, with no collision and zero cumulative environment cost. Raw physical returns are 30.185066414951063 and 30.185057268439166, a difference of -9.1465×10^{-6}. The registered material-loss condition—a new collision, lost success, or at least 5% return loss—is false. The separate numerical no-effect conjunction is also false: although terminal category, step count, and the 10^{-4} return tolerance agree, the maximum paired nominal action-coordinate difference is 0.0014961064, exceeding the precommitted 10^{-5} action tolerance. We retain this failed check and do not relabel the experiment as exact permutation invariance.

The pair completes in 17.67 seconds on CPU with 238 physical controls, 238 actor calls and 240 GST calls, including the native post-terminal autoreset predictions. There is no training or model-API call. This fixed intervention is closed without changing the permutation, case, seed, or tolerances. It provides one bounded no-material-loss control, not a general robustness theorem,

proof of numerical equivalence, or validation of a common security mechanism across systems. Full physical and presentation traces are retained under `runs/gensafenav_person_row_permutation_v1/`.

5.10 Duplicate removal before versus after joint forecasting

R007 tests a restricted, researcher-added tracker-to-forecaster interface. At each eligible GST call, we copy the lowest-index complete observed trajectory into the lowest-index slot with no observed history, using temporary tensors. Original valid histories, physical pedestrians, task, weights, and direct current-state actor inputs are protected. The destination slot need not denote the absence of a physical pedestrian. After the single GST forward, we remove the duplicate output by restoring its invalid mask and future XY values. The native consumer does not use its remaining uncertainty channels, which are not claimed restored. Changes to forecasts for other entities can still reach the actor.

Three complete case-0, seed-42 CPU episodes compare the original inputs, this single duplicate with output removal, and exact removal before GST. The last arm restores the original input tensor before forecasting; it is a controlled reversal of this exact operation, not a general defense. The inherited native evaluation settings are the same as in R006.

Condition	Physical return	Controls	Success	Collision
Original	30.185066415	119	Yes	No
Duplicate, remove output	30.087496792	102	Yes	No
Remove before GST	30.185066415	119	Yes	No

Table 6: R007 complete physical episodes. All three have zero cumulative environment cost. The duplicate arm reaches the goal sooner; its small return decrease is not a material-harm result.

The duplicate arm contains 98 eligible, actually consumed interventions. At the first common eligible call, original history inputs and direct actor current-state fields agree, while forecast XY for original valid entities changes by up to 0.0095983. This is a local influence diagnostic, not a forecast-to-truth error or an attack-success metric. Full-episode return changes by -0.0975696 (-0.3232%), below the precommitted loss threshold of 1.5092533; neither collision nor loss of success occurs. Thus the registered material-harm gate fails. Pre-GST removal reproduces the entire original physical trace, actor observations, resets, forecasts, and outcome exactly, but recovery alone cannot turn the failed harm gate into a positive.

The test completes in 27.11 seconds, with 340 physical controls and 343 GST calls including three post-terminal reset predictions, without API calls or training. We close this fixed operation without changing the source rule, duplicate count, task, or seed. This single reused case establishes neither external attacker access nor a shared three-system security mechanism. The execution snapshot, per-call tensors and complete physical records are in `runs/gensafenav_forecast_track_duplicate_v1/`.

6 Discussion and remaining tests

What the evidence changes. The Cosmos table makes a large first-choice success-harm claim impossible on the already measured support, while preserving the one observed local loss and cost sensitivity. The DINO development bank rejects a worsened-terminal-MSE explanation for its return loss and identifies a scorer mismatch that scalar component reweighting cannot repair on those two actions. The 30-root confirmation establishes lower mean utility after offline oracle

substitution for a fixed bank transferred from development. Its original ranker also trails two fixed-action baselines, making limited selector competence part of the result. The R003 native mean-versus-elites check is negative. The R004 fixed correction supplies an exploratory, deployable-form prediction change, but its single native query has opposite utility directions at H15 and open-loop H30. The subsequent complete native episode has higher corrected cumulative return, so its persistent-loss gate is also negative. These are interface-specific diagnoses. None implies that all attacks fail, that prediction accuracy is irrelevant, or that all world-model agents are objective-limited.

What the confirmation does and does not test. The completed protocol fixes roots, candidate construction, utility, scorer, tie handling, and the oracle comparison before confirmation. It tests the paired utility contrast directly; positive oracle mismatch alone would not have sufficed. The negative interval supports the specified average effect for this bank and root-generating procedure, despite 18 utility ties. The oracle's feature accuracy is guaranteed, while its availability is diagnostic only. The later fixed-bias test avoids evaluation truth as a correction input, but reuses already examined roots and cannot inherit the oracle test's confirmation status. A new confirmation of this correction would require disjoint, prespecified roots and both actual forecast-error and utility measurements. Independent confirmation across task families and native planning interfaces remains unexecuted.

Native planning versus an added ranker. A low-cost fixed-bank diagnostic is useful for isolating the scoring question. Both the DINO two-mean comparison and the 30-root panel use added rankers; a fixed bank that is the actual native selector's support can retain its native interface designation. A native CEM test must record the actual proposal updates and returned mean, evaluate any newly returned executable sequence in the environment, and specify whether its claim concerns a frozen population or a changed full planning process. Holding only the final elites fixed does not reproduce native CEM or include the returned mean automatically. R003 and R004 phase 2 now record and replay native means explicitly. R004's later proposal populations differ between conditions, so its prefix result cannot be attributed solely to a different final ranking on fixed actions. Moreover, H30 open-loop continuation improves while H15 worsens. The separately measured complete episode with native replanning also improves in primary return, despite later first success. These results show why a prefix or open-loop continuation cannot stand in for a complete native episode.

Defenses and generalization. Oracle futures are an offline instrument, not a deployment defense. An eventual scorer-side repair must be fitted on development data and compared with a prespecified forecast-side correction on disjoint roots. Report absolute utilities for each condition, including normal-input behavior, rather than only the gap between conditions. An independent task family is needed before the observed phenomenon can be described as general across tasks or models. The fixed-bank oracle protocol has been executed in full, and two single-root native-query follow-ups and a complete-episode pair are reported separately. A trained-scorer comparison and independent confirmation of the fixed correction are not established by these single-root or reused-root results. Proposed forecast-service integration permissions are assumptions to test, not evidence of an existing native attack surface.

7 Limitations

The empirical evidence remains uneven across systems and interfaces. Cosmos's finite support is complete for a first-choice diagnostic, not for its full closed-loop action space. The DINO confirmation has 30 independently reset roots from one task family and a bank of five development-derived sequences, shared across those roots. This isolates the added scorer's choices but does not sample fresh native CEM proposals or establish full closed-loop performance. Candidate count, mixture grid size, repeated rendering, and factorial cells cannot enlarge the independent sample size. The earlier post-hoc development banks do not enlarge the confirmation sample either. Project-held-out seeds do not establish a holdout from model pretraining. The original ranker's lower mean utility than two fixed candidates limits claims about normally competent planning; a scorer trained and validated on separate roots has not been tested.

R004 phase 1 is an exploratory reuse of the 30-root panel. Its correction improves candidate-averaged error, not every candidate's error, and carries no statistical calibration guarantee. Phase 2 contains one fresh root and one planning query per condition. Its error diagnostic covers only the two returned means, while later CEM proposals differ. The H15 reward and return loss reverses at open-loop H30. Phase 3 completes both native episodes at the same already inspected root and finds higher corrected 99-control return; the preset persistent-loss condition is absent. Later first success and better final success point in different directions and remain secondary to that endpoint. Initial-query MSE improvement does not establish full-episode forecast accuracy, and the full episode is not an independent-root replication. The expanded H15 return criterion was added after launch and before replay outcomes; this timing must not be represented as prelaunch registration. R003's negative result is retained without replacing its root. None of these follow-ups establishes a distinct external security mechanism or a third independent affected system.

The actual-future representation is limited to the model's observation and encoder; oracle-feature scoring need not expose a physical state or an optimal reward representation. Zero oracle error is a constructed manipulation check, and intermediate mixtures may be off the feature manifold. Endpoint accuracy need not measure path accuracy, and terminal proxies need not optimize cumulative return. The prospective primary endpoint is terminal reward and decreases along with cumulative return, so that result is not only a cumulative-versus-terminal metric distinction. Success is sparse (2/30 versus 0/30) and descriptive; the dense-reward result is not a broad claim of induced task failure.

The interface bounds depend on complete recorded support, matched continuation policies, and the stated outcome definition. A change that produces a new mean action, changes later controls, or changes an environment's hidden-state distribution is outside them. One realized stochastic branch per candidate does not provide a bound on its expected return. Full mediator identification, attacker reachability, and deployment permissions require additional evidence. The archival source includes systems with incompatible, retired, or incomplete protocols; these cannot be pooled into a seven-system replication claim.

8 Conclusion

Fixed-support utility tables and actual-future scoring answer two questions that forecast error alone cannot: how much the measured decision interface can change real outcomes, and whether its frozen objective would choose well given the observed futures. The Cosmos cache bounds first-choice success loss on its measured support. A completed 30-root DINO confirmation shows that offline actual-future substitution lowers mean terminal reward by 1.542393 in an added five-action ranker,

while simple fixed choices outperform its original scoring. A fixed additive correction reproduces a smaller exploratory effect without evaluation futures as correction inputs. Its native single-query comparison reduces H15 reward and return but improves both under H30 open-loop continuation. Complete native replanning at that same root increases 99-control return by 12.738701, while delaying first success; the preset persistent-return-loss gate is negative, as is the native elite-mean gate. Together these findings support a reproducible diagnostic study of a known prediction–control mismatch, with an explicit negative full-episode result for the proposed loss extension. They establish neither a new security mechanism nor independent cross-task generality, full-episode forecast-fidelity improvement, or a validated repair.

References

[1] Anonymous. Latent metrics for planning with world models, 2026. ICLR 2027 conference submission, under review. Official abstract and metadata verified; full text not accessed.

[2] Logan Mondal Bhamidipaty, Esmeralda S. Whitammer, David Abel, Mykel J. Kochenderfer, and Subramanian Ramamoorthy. Imperfect world models are exploitable, 2026. Sections 1 and 2.2 checked in version 1; author metadata checked against the arXiv abstract page.

[3] Morgan Byrd, Robert Wright, and Sehoon Ha. CF-JEPA: Improving robustness of JEPA world models via controllability factorization, 2026.

[4] Adam N. Elmachtoub and Paul Grigas. Smart "predict, then optimize". *Management Science*, 68(1):9–26, 2022.

[5] Amir-Massoud Farahmand, Andre Barreto, and Daniel Nikovski. Value-aware loss function for model-based reinforcement learning. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pages 1486–1494, 2017.

[6] Christopher Grimm, Andre Barreto, Satinder Singh, and David Silver. The value equivalence principle for model-based reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

[7] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

[8] Yibin Hu, Xiaolin Sun, and Zizhan Zheng. Stealthy world model manipulation via data poisoning, 2026. Version 1, appendix E checked for native proposal and evaluation roles.

[9] Moo Jin Kim, Yihuai Gao, Tsung-Yi Lin, Yen-Chen Lin, Yunhao Ge, Grace Lam, Percy Liang, Shuran Song, Ming-Yu Liu, Chelsea Finn, and Jinwei Gu. Cosmos policy: Fine-tuning video models for visuomotor control and planning, 2026.

[10] Nathan Lambert, Brandon Amos, Omry Yadan, and Roberto Calandra. Objective mismatch in model-based reinforcement learning. In *Proceedings of the 2nd Conference on Learning for Dynamics and Control*, volume 120 of *Proceedings of Machine Learning Research*, pages 761–770. PMLR, 2020.

[11] Jiazhuo Li, Yiming Fei, Zhiruo Zhou, and Heikichi Hayashi. Dueling world models: Advantage-style action channels for common-mode distractor rejection, 2026.

[12] Alexander Pan, Kush Bhatia, and Jacob Steinhardt. The effects of reward misspecification: Mapping and mitigating misaligned models. In *International Conference on Learning Representations*, 2022.

[13] Basile Terver, Tsung-Yen Yang, Jean Ponce, Adrien Bardes, and Yann LeCun. What drives success in physical planning with joint-embedding predictive world models?, 2025.

[14] Huining Yuan, Hongkun Dou, Xingyu Jiang, and Yue Deng. Task-aware world model learning with meta weighting via bi-level optimization. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[15] Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World models on pre-trained visual features enable zero-shot planning, 2024. Version 2 revised February 2025.

A Evidence ledger and reproduction

Paths below are relative to the repository root. The paper build uses cached numerical analyses and does not call a model API. The repository's earlier knowledge-base update and this manuscript may be newer than the public GitHub baseline; the artifact manifest records content hashes rather than assuming that a branch name identifies the current working tree.

Finite-support aggregation
`scripts/compute_decision_leverage_panel.py`, with generated `reports/decision_leverage_paper/panel_*` outputs. Includes explicit eligibility and exclusion reasons; incomplete systems are not assigned zero leverage.

Cosmos actual branch outcomes and bounds
`runs/cosmos_first_candidate_utility_recovery_analysis_v1/analysis.json`
`reports/cosmos_first_choice_identifiability_bounds_20261015.json`
`reports/cosmos_first_selection_mediation_20261015.json`.

DINO measured H30 countercheck
`reports/dino_first_query_terminal_truth_decision_20261015.json`
`scripts/analyze_dino_first_query_objective_decomposition.py`.

New objective-family calculation
`scripts/analyze_decision_objective_reweighting.py`
`reports/decision_leverage_paper/objective_reweighting.json`.

Cosmos actual-future and modality controls
`reports/cosmos_first_query_truth_decision_20261015.json`
`reports/cosmos_future_modality_factorial_decision_20261016.json`.

Completed prospective DINO confirmation
`configs/decision_leverage_dino_e1.json`
`reports/decision_leverage_paper/prospective_dino.json`
`runs/decision_leverage_dino_cpu_confirmatory/`. The design configuration records the planned protocol and retains a prelaunch execution-status field. The completed analysis JSON is the execution evidence: all 30 registered roots are included, calibration roots are excluded, and its validation record reports source bindings and verified raw artifact hashes. It contains the paired bootstrap

summaries, all mixture arms, baselines, activity receipt, and interpretation limits. Optional scorer-training roots remain unexecuted. The later native-query follow-ups below have their own protocols and do not enlarge this confirmation sample.

Negative native elite-mean gate, R003
`runs/cem_elite_mean_utility_gate_v1/summary.json`. Records the unchanged native query, all eleven real prefix outcomes, exact mean/elite consistency checks, and the absent primary condition.

Fixed residual correction, R004 phase 1
`runs/forecast_residual_calibration_v1/fit_commitment.json`
`runs/forecast_residual_calibration_v1/summary.json`
`reports/forecast_residual_calibration.json`. The saved bias predates evaluation reads. The full report retains all 30 reused roots and the stronger fixed-action baselines; inference is descriptive.

Native residual-correction query, R004 phase 2
`runs/native_forecast_calibration_gate_v1/summary.json`
`runs/native_forecast_calibration_gate_v1/interpretation.json`. The interpretation records the after-launch, before-outcome criterion addition and the opposite H15/H30 utility directions. These files are not evidence of a complete episode; the subsequent episode experiment is recorded separately.

Complete native episodes, R004 phase 3
`runs/native_forecast_calibration_episode_v1/summary.json`
`runs/native_forecast_calibration_episode_v1/raw/result.json`
`runs/native_forecast_calibration_episode_v1/calibrated/result.json`. Each arm also retains its 99-step `trace.jsonl`. The first query exactly repeats phase 2 as an implementation check. The complete-episode cumulative-return loss condition is false; no full-episode forecast-error claim is recorded. The figure is `reports/decision_leverage_paper/native_calibration_episode.pdf`, with its companion source ledger.

B Utility and interface reporting requirements

System / interface	Utility	Roots	Range	Baseline	Harm cap
Cosmos	Success count	30	[28, 30]	29	1
Cosmos	Capped steps	30	[3230, 3646]	3477	169
DINO-WM	30-step return	1	[24.23, 110.152]	109.057	84.827
DINO-WM	Terminal reward	1	[0.481, 7.989]	7.699	7.218
DINO-WM	30-step success	1	[0, 0]	0	0
WorldAgen	3-step success	3	[0, 0]	0	0
WorldAgen	3-step reward	3	[0, 0]	0	0

Table 7: Generated archival finite-support panel. DINO uses the seed1 five-action development bank and an added ranker. WorldAgen supplies three short roots from one episode and no forecast-driven native selector; its zero three-step task reward is a local metric observation, not evidence of full-task robustness. Systems lacking eligible real continuation tables are excluded rather than assigned zero bounds. The separate completed 30-root DINO confirmation is reported in Table 2.

Every new record should identify the originating episode, task, root state, environment and wrapper versions, original task Goal, model checkpoint, candidate bytes or content hash, noise and

continuation settings, native or added selector status, forecast horizon, execution horizon, and tie rule. Report success, dense return, terminal reward or distance, and capped cost only where their definitions are available. A missing utility is missing, not zero. Preserve the predeclared primary measure when secondary metrics disagree, and show all retained measures so that harm cannot be manufactured by choosing a favorable metric after inspecting the result.